Type I

Integrals of the form $\int \frac{dx}{a^2 \pm x^2}$, $\int \frac{dx}{x^2 - a^2}$, $\int \frac{dx}{\sqrt{a^2 \pm x^2}}$, $\int \frac{dx}{\sqrt{x^2 - a^2}}$

(i) $$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + c$$

(ii) $$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + c$$

(iii) $$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + c$$

(iv) $$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + c$$

(v) $$\int \frac{dx}{\sqrt{x^2 - a^2}} = \log \left| x + \sqrt{x^2 - a^2} \right| + c$$

(vi) $$\int \frac{dx}{\sqrt{x^2 + a^2}} = \log \left| x + \sqrt{x^2 + a^2} \right| + c$$

Proof

(i) Let $I = \int \frac{dx}{a^2 - x^2}$.

$$= \int \frac{dx}{(a-x)(a+x)}$$

$$= \frac{1}{2a} \int \left[\frac{1}{a+x} + \frac{1}{a-x} \right] dx \qquad \left\{ \begin{matrix} \text{Resolving into} \\ \text{partial fractions} \end{matrix} \right.$$

$$= \frac{1}{2a} \left[\log |a+x| - \log |a-x| \right] + c$$

$$= \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + c$$

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + c$$

(ii) Let $I = \int \frac{dx}{x^2 - a^2}$.

$$= \int \frac{dx}{(x-a)(x+a)}$$

$$= \frac{1}{2a} \int \left[\frac{1}{x-a} - \frac{1}{x+a} \right] dx \qquad \left\{ \begin{matrix} \text{Resolving into} \\ \text{partial fractions} \end{matrix} \right.$$

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